

Exp no:8

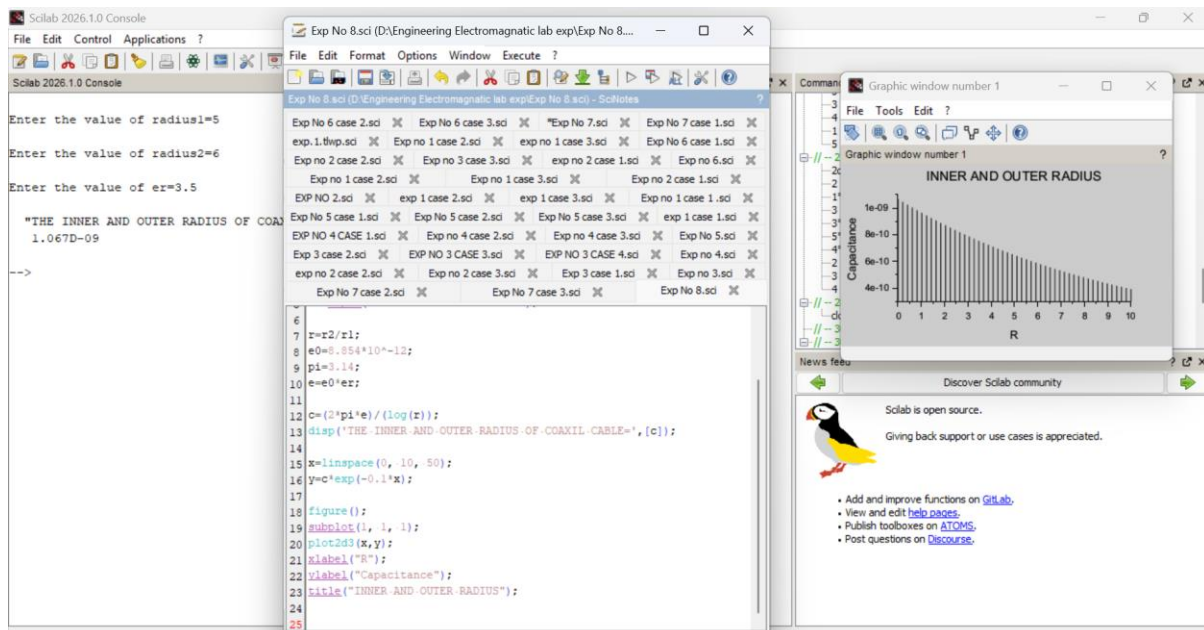
Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Aim:

To write program for Capacitance of co-axial cable per unit length using SCILAB.

Software required:

- SCILAB version 6.0.1



EXP NO :- 08

$$r_1 = 5 \text{ cm}$$

$$r_2 = 6 \text{ cm}$$

$$\epsilon_r = 3.5$$

$$C = \frac{2\pi \epsilon_0 \epsilon_r L}{\ln(r_2/r_1)}$$

$$2\pi \epsilon_0 \epsilon_r L$$

$$= 2 \times 3.14 \times 8.854 \times 10^{-12} \times 3.5 \times 1$$

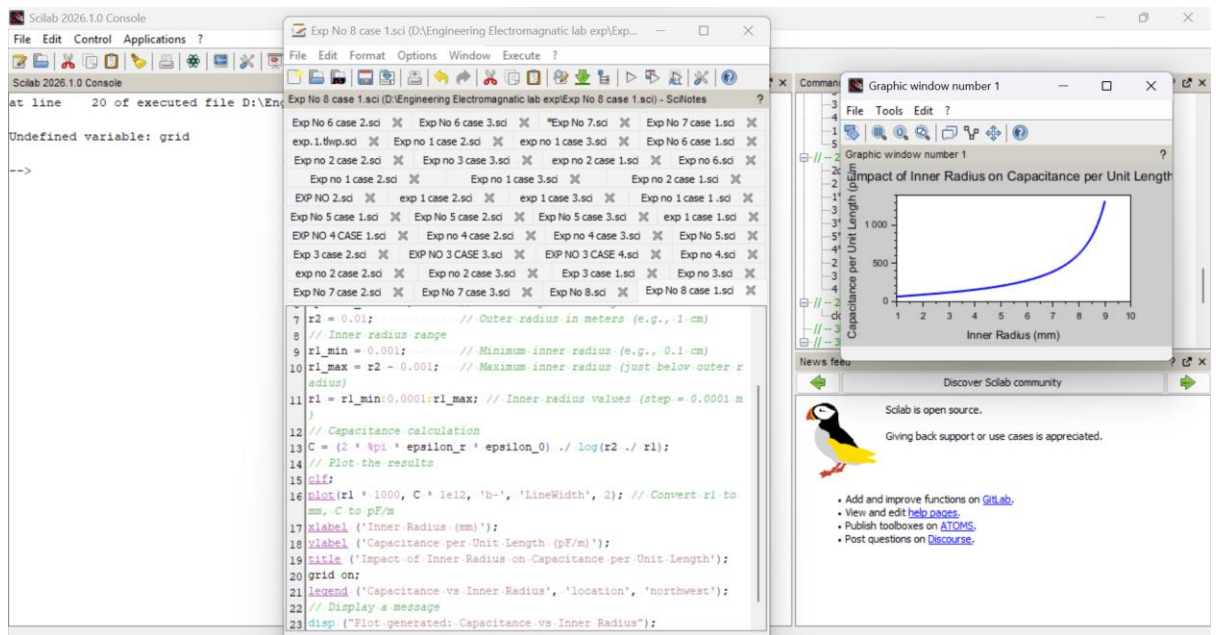
$$= 1.947 \times 10^{-10}$$

$$\ln(r_2/r_1) = 0.1823$$

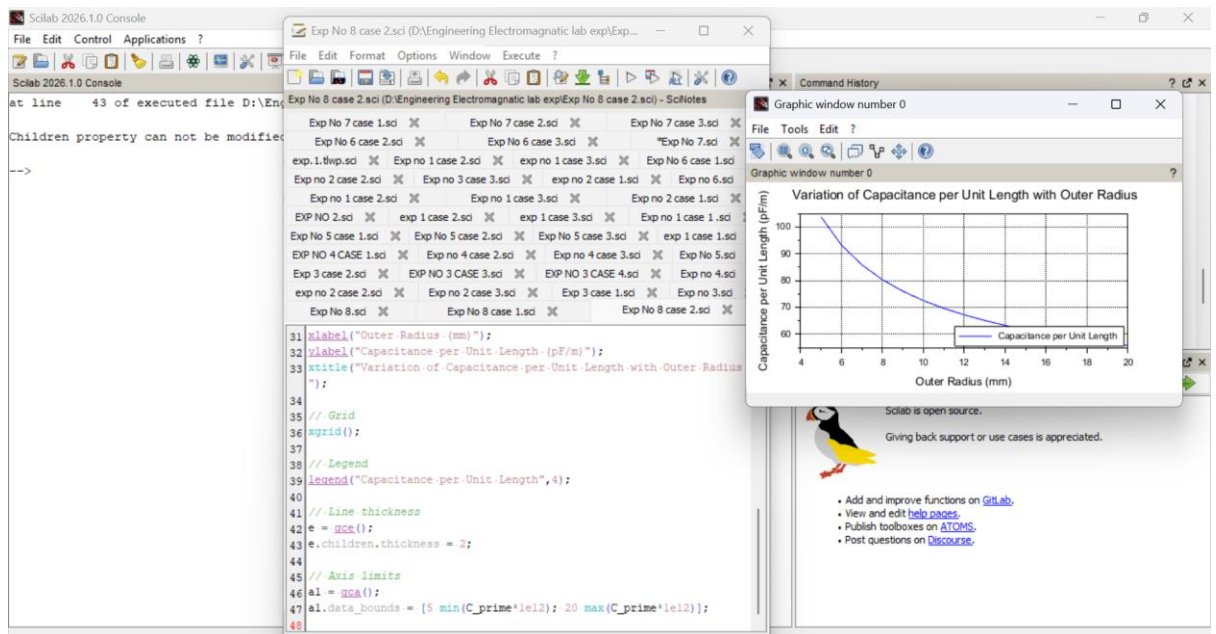
$$C = \frac{1.947 \times 10^{-10}}{0.1823}$$

$$= 1.068 \times 10^{-9} \text{ F}$$

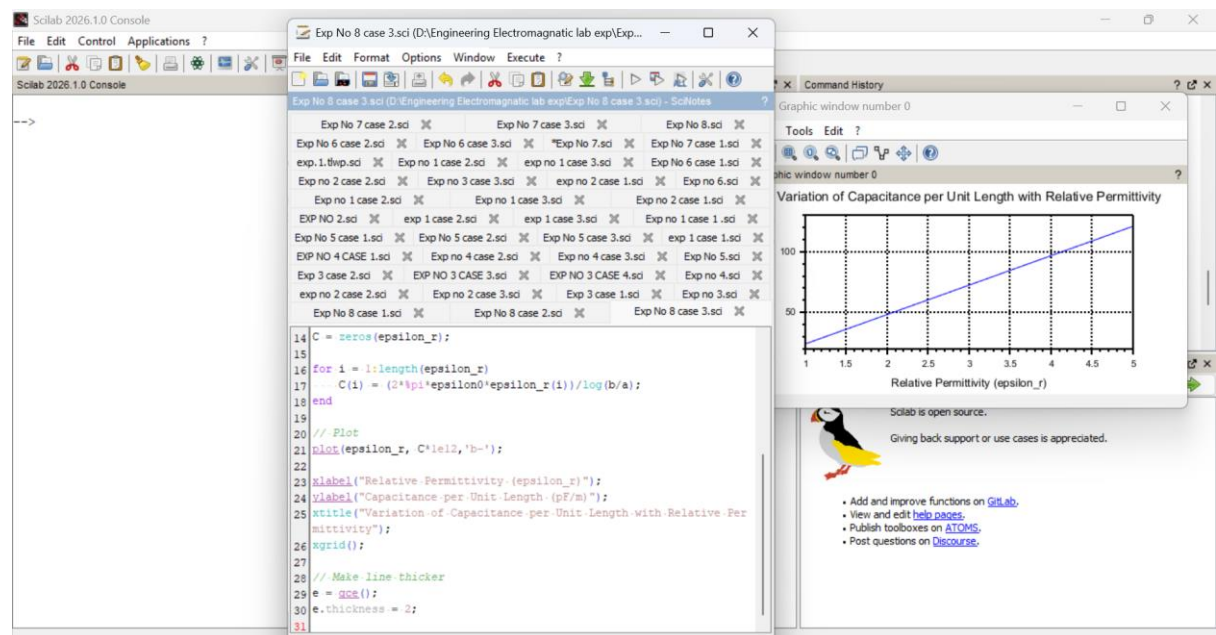
$$C = 1.067 \times 10^{-9} \text{ C}$$



CASE STUDY 2: Variation with Outer Radius



CASE STUDY 3: Variation with Relative Permittivity



Result :

Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.